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ECU replacement with odometer

Abstract

A computer includes a processor and a memory, and the memory stores instructions executable by the processor to receive a first data block from a server remote from the computer and remote from a vehicle; while the computer is installed on board the vehicle, receive a second data block from an electronic control unit (ECU) on the vehicle; compare the first data block and the second data block; upon identifying a match between the first data block and the second data block, replace a stored odometer value on the computer with a first odometer value; and upon identifying a mismatch between the first data block and the second data block, maintain the stored odometer value on the computer. The first data block includes the first odometer value, and the second data block includes a second odometer value.

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Background/Summary

BACKGROUND

(1) An odometer is an instrument for measuring the distance traveled by a vehicle. The value of the odometer may be used as an estimate for the wear and tear of a vehicle, for example, when selling a used vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a block diagram of an example system for replacing an electronic control unit (ECU) including an odometer in a vehicle before replacement of the ECU.
- (2) FIG. 1B is a block diagram of the system after the replacement of the ECU.
- (3) FIG. 2A is a diagram of an example third data block containing an odometer value.
- (4) FIG. 2B is a diagram of an example second data block containing an odometer value.
- (5) FIG. 2C is a diagram of an example first data block containing an odometer value.
- (6) FIGS. 3A-C are collectively a sequence diagram of an example sequence for replacing the ECU.
- (7) FIG. 4 is a flowchart of an example process for the ECU to replace the ECU.
- (8) FIG. 5 is a flowchart of an example process for a second ECU to replace the ECU.
- (9) FIG. 6 is a flowchart of an example process for a remote server to replace the ECU.

DETAILED DESCRIPTION

(10) This disclosure provides techniques for updating an odometer value in a secure manner that may prevent tampering, e.g., when installing an electronic control unit (ECU) containing the odometer value in a used vehicle. While on board a vehicle, an original first ECU may back up a third data block including a third odometer value on a second ECU on board the vehicle. During replacement, the original first ECU or the second ECU may send the third data block to the service center, which forwards the third data block to a remote server. A replacement first ECU may receive a first data block from the remote server, receive a second data block from the second ECU, and compare the first data block and the second data block. If both the first data block and the second data block were derived from the third data block, then the first data block and the second data block should have matching parts. Upon identifying a match between the first data block and the second data block, the replacement first ECU replaces a stored odometer value on the replacement first ECU with a first odometer value from the first data block. Upon identifying a mismatch between the first data block and the second data block, the replacement first ECU

maintains the stored odometer value on the replacement first ECU. The replacement first ECU may thus detect attempted tampering with the comparison between the first data block from the remote server and the second data block backed up on board the vehicle. The replacement first ECU may thus take on an odometer value that is accurate to the vehicle, even if that odometer value is lower than the odometer value already on the replacement first ECU (e.g., if the replacement first ECU is used). The techniques herein facilitate replacement of an ECU that serves as the odometer of the vehicle.

(11) A computer includes a processor and a memory, and the memory stores instructions executable by the processor to receive a first data block from a server remote from the computer and remote from a vehicle; while the computer is installed on board the vehicle, receive a second data block from an electronic control unit (ECU) on the vehicle; compare the first data block and the second data block; upon identifying a match between the first data block and the second data block, replace a stored odometer value on the computer with the first odometer value; and upon identifying a mismatch between the first data block and the second data block, maintain the stored odometer value on the computer. The first data block includes a first odometer value, and the second data block includes a second odometer value.

(12) In an example, the instructions may further include instructions to increment the stored odometer value with distance traveled by the vehicle.

(13) In an example, the instructions may further include instructions to transmit a third data block to the ECU, the third data block including the stored odometer value. In a further example, the instructions may further include instructions to periodically transmit the third data block to the ECU.

(14) In another further example, the instructions may further include instructions to, upon the vehicle traveling a preset distance since a previous transmission of the third data block to the ECU, transmit the third data block to the ECU.

(15) In another further example, the instructions may further include instructions to calculate a security feature and apply the security feature to the third data block before transmitting the third data block to the ECU.

(16) In an example, the first odometer value may be less than the stored odometer value.

(17) In an example, the match may be an equality between the first odometer value and the second odometer value, and the mismatch may be an inequality between the first odometer value and the second odometer value.

(18) In an example, the first data block may include a first vehicle identification number (VIN), the second data block may include a second VIN, the match may be an equality between the first VIN and the second VIN, and the mismatch may be an inequality between the first VIN and the second VIN. In a further example, the instructions may further include instructions to, upon the first VIN not matching a third VIN of the vehicle, maintain the stored odometer value on the computer. In a yet further example, the instructions may further include instructions to determine the third VIN from a vehicle network of the vehicle.

(19) In another further example, the instructions may further include instructions to, upon the second VIN not matching a third VIN of the vehicle, maintain the stored odometer value on the computer.

(20) In an example, the instructions may further include instructions to, upon identifying the mismatch between the first data block and the second data block, instruct a component of the vehicle to disable an operation of the component.

(21) In an example, the instructions may further include instructions to, upon identifying the mismatch between the first data block and the second data block, output a request for a fourth data block, the fourth data block previously downloaded from the vehicle, the fourth data block including a fourth odometer value. In a further example, the instructions may further include instructions to compare the first data block and the fourth data block; upon identifying a second

match between the first data block and the fourth data block, replace the stored odometer value on the computer with the first odometer value; and upon identifying a second mismatch between the first data block and the fourth data block, maintain the stored odometer value on the computer. In a yet further example, the first data block may include a first security feature, the fourth data block may include a fourth security feature, the second match may be an equality between the first security feature and the fourth security feature, and the second mismatch may be an inequality between the first security feature and the fourth security feature. In a still yet further example, the first security feature may be calculated from the first odometer value, and the fourth security feature may be calculated from the fourth odometer value.

(22) In another further example, the instructions may further include instructions to, upon failing to receive the fourth data block, maintain the stored odometer value on the computer.

(23) In an example, the instructions may further include instructions to decrypt the first data block, and decrypt the second data block.

(24) A method includes receiving a first data block from a server remote from a computer and remote from a vehicle; receiving a second data block from an electronic control unit (ECU) on the vehicle by the computer; comparing the first data block and the second data block; upon identifying a match between the first data block and the second data block, replacing a stored odometer value on the computer with the first odometer value; and upon identifying a mismatch between the first data block and the second data block, maintaining the stored odometer value on the computer. The first data block includes a first odometer value, and the second data block includes a second odometer value. The computer receives the second data block from the ECU while the computer is installed on board the vehicle.

(25) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a first electronic control unit (ECU) **105** includes a processor and a memory, and the memory stores instructions executable by the processor to receive a first data block **250** from a remote server **125** remote from the first ECU **105** and remote from a vehicle **100**; while the first ECU **105** is installed on board the vehicle **100**, receive a second data block **225** from a second ECU **110** on the vehicle **100**; compare the first data block **250** and the second data block **225**; upon identifying a match between the first data block **250** and the second data block **225**, replace a stored odometer value **205** on the first ECU **105** with a first odometer value **255**; and upon identifying a mismatch between the first data block **250** and the second data block **225**, maintain the stored odometer value **205** on the first ECU **105**. The first data block **250** includes the first odometer value **255**, and the second data block **225** includes a second odometer value **230**.

(26) With reference to FIGS. **1A-B**, the vehicle **100** may be any passenger or commercial automobile such as a car, a truck, a sport utility vehicle, a crossover, a van, a minivan, a taxi, a bus, etc.

(27) The first ECU **105** is a microprocessor-based computing device, e.g., a generic computing device including a processor and a memory, an electronic controller or the like, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a combination of the foregoing, etc. Typically, a hardware description language such as VHDL (VHSIC (Very High Speed Integrated Circuit) Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. The first ECU **105** can thus include a processor, a memory, etc. The memory of the first ECU **105** can include media for storing instructions executable by the processor as well as for electronically storing data and/or databases, and/or the first ECU **105** can include structures such as the foregoing by which programming is provided.

(28) This disclosure pertains to techniques for replacing the first ECU **105** on board the vehicle

100. The first ECU **105** on board the vehicle **100** before replacement will be referred to as the original first ECU **105a**, and the first ECU **105** on board the vehicle **100** after replacement will be referred to as the replacement first ECU **105b**. FIG. 1A shows the vehicle **100** before replacement, with the original first ECU **105a** installed on board the vehicle **100**. FIG. 1B shows the vehicle **100** after replacement, with the replacement first ECU **105b** installed on board the vehicle **100**. The original first ECU **105a** and the replacement first ECU **105b** may both include some or all of the programming described herein.

(29) When installed on board the vehicle **100**, the first ECU **105** may transmit and receive data through a vehicle network **115** such as a controller area network (CAN) bus, Ethernet, WiFi, Local Interconnect Network (LIN), onboard diagnostics connector (OBD-II), and/or by any other wired or wireless communications network. The first ECU **105** may be communicatively coupled to the second ECU **110**, a transceiver **120**, and other components via the vehicle network **115**.

(30) The second ECU **110** is a microprocessor-based computing device, e.g., a generic computing device including a processor and a memory, an electronic controller or the like, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a combination of the foregoing, etc. Typically, a hardware description language such as VHDL (VHSIC (Very High Speed Integrated Circuit) Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. The second ECU **110** can thus include a processor, a memory, etc. The memory of the second ECU **110** can include media for storing instructions executable by the processor as well as for electronically storing data and/or databases, and/or the second ECU **110** can include structures such as the foregoing by which programming is provided. The second ECU **110** may be a gateway module of the vehicle **100**. The gateway module is an ECU that connects and transmits data between buses of different domains of the vehicle network **115**, e.g., the CAN bus, Ethernet, LIN, OBD-II, etc., which can have different baud rates.

(31) The transceiver **120** may be adapted to transmit signals wirelessly through any suitable wireless communication protocol, such as cellular, Bluetooth®, Bluetooth® Low Energy (BLE), ultra-wideband (UWB), WiFi, IEEE 802.11a/b/g/p, cellular-V2X (CV2X), Dedicated Short-Range Communications (DSRC), other RF (radio frequency) communications, etc. The transceiver **120** may be adapted to communicate with the remote server **125** and/or other servers remote from the vehicle **100**. For example, a server may be associated with another vehicle (e.g., V2V communications), an infrastructure component (e.g., V2I communications), an emergency responder, a mobile device associated with the owner of the vehicle **100**, etc. The transceiver **120** may be one device or may include a separate transmitter and receiver.

(32) The replacement of the first ECU **105** may take place at a service center **130**. The service center **130** may be a facility such as a repair shop or dealership. The service center **130** may communicate with the replacement first ECU **105b** before installation or the original first ECU **105a** after replacement by plugging equipment directly into the first ECU **105**. The service center **130** may communicate with components on board the vehicle **100** via the transceiver **120** or by plugging equipment directly into the vehicle network **115**, e.g., via an OBD-II port.

(33) The transceiver **120** and/or the service center **130** may be connected to the remote server **125** via a network **135**. The network **135** represents one or more mechanisms by which the transceiver **120** or other components may communicate with the remote server **125**. Accordingly, the network **135** may be one or more of various wired or wireless communication mechanisms, including any desired combination of wired (e.g., cable and fiber) and/or wireless (e.g., cellular, wireless, satellite, microwave, and radio frequency) communication mechanisms and any desired network topology (or topologies when multiple communication mechanisms are utilized). Exemplary communication networks include wireless communication networks (e.g., using Bluetooth, IEEE

802.11, etc.), local area networks (LAN) and/or wide area networks (WAN), including the Internet, providing data communication services.

(34) The remote server **125** is remote from the vehicle **100**, that is, distinct and spaced from the vehicle **100**. The remote server **125** is located outside the vehicle **100**. The remote server **125** is also remote from the replacement first ECU **105b** before the replacement first ECU **105b** is installed on the vehicle **100**, as well as remote from the service center **130**. The remote server **125** may be associated with, e.g., a manufacturer of the vehicle **100**.

(35) With reference to FIG. 2A, the third data block **200** is a data block outputted by the first ECU **105**. The third data block **200** may include the stored odometer value **205**, a third vehicle identification number (VIN) **210**, a third part number **215**, and a third security feature **220**.

(36) The memory of the first ECU **105** stores the stored odometer value **205**. The first ECU **105** may be programmed to generate the stored odometer value **205** by incrementing the stored odometer value **205** with distance traveled by the vehicle **100** (starting at zero, e.g., when the vehicle **100** has just been assembled or if the first ECU **105** is new). In other words, the first ECU **105** may operate as an odometer of the vehicle **100**. As the first ECU **105** receives data over the vehicle network **115** indicating that the vehicle **100** is traveling, the first ECU **105** may increase the stored odometer value **205** accordingly. For example, the first ECU **105** may receive data indicating a number of revolutions of the wheels of the vehicle **100** and multiply the number of revolutions by a circumference of the tires of the vehicle **100**, e.g., $\Delta d = \pi DN$, in which Δd is a change in the stored odometer value **205**, D is a diameter of the tires, and N is a number of revolutions of the tires, e.g., an average of the numbers of revolutions of each of the tires.

(37) A VIN is a number encoding specific information about a specific vehicle, e.g., make and model, manufacturer, year of manufacture, plant of manufacture, etc. The VIN is unique to one vehicle and thus identifies the vehicle **100**. For example, under National Highway Traffic Safety Administration (NHTSA) regulations, the VIN is seventeen characters long and is formed of letters and numbers. The third VIN **210** may be stored in the memory of the first ECU **105**.

(38) A part number is a number uniquely identifying a component of the vehicle **100**. The part number may be unique to the component among components of the same type. For example, the third part number **215** may identify the first ECU **105**. The third part number **215** may be unique among ECUs of a same model as the first ECU **105**. The third part number **215** may be stored in the memory of the first ECU **105**.

(39) The first ECU **105** may be programmed to calculate the third security feature **220**. The third security feature **220** may be, e.g., a hash value (as shown in the Figures), a digital signature, a cipher-based message authentication code (CMAC), etc. A hash value is a numeric value of fixed length that uniquely identifies data. The first ECU **105** may calculate the hash value by applying a hashing function to one or more other parts of the third data block **200**, e.g., the stored odometer value **205**, the third VIN **210**, and/or the third part number **215**. A hash function maps data of arbitrary size to a hash value of fixed size. The hash function may be any suitably secure hash function, e.g., SHA-256. A digital signature verifies a sender of a message and is determined from a private key and one or more other parts of the third data block **200**. The digital signature may be verified using a public key. A CMAC is a message authentication code constructed from a block cipher. The CMAC may be determined from a secret key and one or more other parts of the third data block **200**. The CMAC may be verified with the secret key. The third security feature **220** is then applied to the third data block **200**, e.g., appended to the third data block **200**, before the third data block **200** is transmitted to another entity.

(40) The first ECU **105** may be programmed to transmit the third data block **200** to the second ECU **110**, e.g., over the vehicle network **115** while the first ECU **105** is installed on board the vehicle **100**. For example, the first ECU **105** may periodically transmit the third data block **200** to the second ECU **110**, i.e., transmit the third data block **200** repeatedly at regular periods. The periods may be measured in distance traveled by the vehicle **100** or in time, i.e., the period may be a preset

distance or preset time. For example, the first ECU **105** may transmit the third data block **200** to the second ECU **110** upon the vehicle **100** traveling the preset distance since a previous transmission of the third data block **200** to the second ECU **110**. The preset distance may be, e.g., ten miles. The first ECU **105** may track progress to the preset distance using the stored odometer value **205**.

(41) The first ECU **105** may encrypt the third data block **200** before transmitting the third data block **200** to the second ECU **110**, e.g., using any suitable encryption algorithm. The second ECU **110**, as well as the other ECUs on board the vehicle **100**, may lack the decryption key. The third data block **200** may therefore be inaccessible to someone without the decryption key despite the third data block **200** residing on the second ECU **110**. The decryption key may be stored on the remote server **125**.

(42) With reference to FIG. 2B, the second data block **225** is a data block outputted by the second ECU **110**. The second data block **225** may be the most recent third data block **200** received by the second ECU **110**. The second ECU **110** may be programmed to transmit the second data block **225** to an entity upon receiving a request for the second data block **225** from the entity. The second data block **225** may include the second odometer value **230**, a second VIN **235**, a second part number **240**, and a second security feature **245**. If the most recent third data block **200** received by the second ECU **110** came from the original first ECU **105a** and the replacement first ECU **105b** has been installed, then the second part number **240** will be different from the third part number **215** of the third data block **200** outputted by the replacement first ECU **105b**.

(43) With reference to FIG. 2C, the first data block **250** is a data block outputted by the remote server **125**. The first data block **250** includes the first odometer value **255**, a first VIN **260**, a first part number **265**, a third-block security feature **275**, and a first security feature **270**. The remote server **125** may be programmed to receive the third data block **200** or second data block **225**, decrypt the third data block **200** or second data block **225**, and generate the first data block **250** from the decrypted third data block **200** or second data block **225**. For example, the service center **130** may download the third data block **200** or second data block **225**, and the remote server **125** may receive the third data block **200** or second data block **225** from the service center **130** over the network **135**. The service center **130** may send the third data block **200** if the original first ECU **105a** is operational and the second data block **225** if the original first ECU **105a** is not operational. The remote server **125** decrypts the third data block **200** or second data block **225** using the decryption key stored at the remote server **125**. The remote server **125** may use the stored odometer value **205** or the second odometer value **230** as the first odometer value **255**, the second VIN **235** or third VIN **210** as the first VIN **260**, the second part number **240** or third part number **215** as the first part number **265**, and the second security feature **245** or third security feature **220** as the third-block security feature **275**. The service center **130** may transmit the VIN of the vehicle **100** to use as the first VIN **260**. The remote server **125** may be programmed to calculate the first security feature **270**, e.g., by applying the same algorithm as used by the first ECU **105** to determine the third security feature **220** to one or more parts of the first data block **250**, e.g., the equivalent parts as the first ECU **105** when calculating the third security feature **220**, e.g., the first odometer value **255**, the first VIN **260**, and/or the first part number **265**, as well as possibly the third-block security feature **275**.

(44) The first ECU **105** upon installation, i.e., the replacement first ECU **105b**, may be programmed to compare the first data block **250** received from the remote server **125** and the second data block **225** received from the second ECU **110**. The first ECU **105** identifies either a match or a mismatch between equivalent portions of the first data block **250** and the second data block **225**. A match means that the portions are equal or identical, and a mismatch means that the portions are unequal or different. For example, the match may be an equality between the first security feature **270** or third-block security feature **275** and the second security feature **245**. The first ECU **105** may also validate the security features **245**, **270**, **275** upon receiving the respective data blocks **225**, **250**, before comparing the data blocks **225**, **250**. For another example, the match

may be an equality between the first odometer value **255** and the second odometer value **230**, and the mismatch may be an inequality between the first odometer value **255** and the second odometer value **230**. For another example, the match may be an equality between the first VIN **260** and the second VIN **235**, and the mismatch may be an inequality between the first VIN **260** and the second VIN **235**. The first ECU **105** may also compare the first VIN **260** and the second VIN **235** to a VIN received over the vehicle network **115**.

(45) The first ECU **105** may be programmed to, upon identifying the mismatch between the first data block **250** and the second data block **225**, output a request for a fourth data block. The fourth data block may be previously downloaded from the vehicle **100**. For example, the fourth data may be the copy of the third data block **200** or second data block **225** downloaded at the service center **130**, and which was sent to the remote server **125**. The service center **130** may transmit the fourth data block to the first ECU **105** upon receiving the request for the fourth data block. The fourth data block may include a fourth odometer value, a fourth VIN, a fourth part number, and a fourth security feature, as described above for the equivalent portions of the second data block **225** and third data block **200**.

(46) The first ECU **105** may be programmed to compare the first data block **250** and the fourth data block. The first ECU **105** identifies either a match or a mismatch between equivalent portions of the first data block **250** and the fourth data block (referred to hereinafter as a second match and a second mismatch to differentiate from the comparison between the first data block **250** and second data block **225** described above). For example, the second match may be an equality between the first security feature **270** or third-block security feature **275** and the fourth security feature, and the second mismatch may be an inequality between the first security feature **270** or third-block security feature **275** and the fourth security feature. The first ECU **105** may also validate the security features **270**, **275** from the first data block **250** upon receiving the first data block **250** and the fourth security feature upon receiving the fourth data block, before comparing the first data block **225** and the fourth data block.

(47) The first ECU **105** may be programmed to replace the stored odometer value **205** on the first ECU **105** with the first odometer value **255** upon either identifying the match between the first data block **250** and the second data block **225** or identifying the second match between the first data block **250** and the fourth data block. The first ECU **105** may be programmed to replace the stored odometer value **205** with the first odometer value **255** even if the first odometer value **255** is less than the stored odometer value **205**.

(48) The first ECU **105** may be programmed to maintain the stored odometer value **205** on the first ECU **105** upon identifying the mismatch and either identifying the second mismatch or failing to receive the fourth data block. The first ECU **105** may thus prevent an unauthorized decrease of the stored odometer value **205**.

(49) The first ECU **105** may be programmed to, upon the same conditions as maintaining the stored odometer value **205**, instruct a component of the vehicle **100** to disable an operation of the component. In other words, the first ECU **105** instructs the component to disable the operation of the component upon identifying the mismatch and either identifying the second mismatch or failing to receive the fourth data block. For example, the component may be an infotainment center, and the operation may be, e.g., navigation and/or media input. Thus, the first ECU **105** may instruct the infotainment center to disable navigation or audio entertainment functionality. The infotainment center may be chosen to make operating the vehicle **100** less desirable while not impairing the operation of the vehicle **100**.

(50) FIGS. 3A-C are collectively a sequence diagram illustrating an example sequence **300** of steps for replacing the first ECU **105**. The sequence **300** may be performed by the original first ECU **105a**, the replacement first ECU **105b**, the second ECU **110**, the service center **130**, and the remote server **125**. The memories of the original first ECU **105a**, the replacement first ECU **105b**, the second ECU **110**, the service center **130**, and the remote server **125** may store executable

instructions for performing the respective steps of the sequence **300** and/or programming can be implemented in structures such as mentioned above. The steps at the service center **130** may be performed manually by a technician at the service center **130**.

(51) In a step **302**, before the replacement and during regular operation of the vehicle **100**, the original first ECU **105a** encrypts and transmits the third data block **200** to the second ECU **110**, as described above.

(52) Next, the vehicle **100** may be brought in to the service center **130** for replacement of the original first ECU **105a** with the replacement first ECU **105b**. An alternative box **304** includes a first alternative, that the original first ECU **105a** is still operational, and a second alternative, that the original first ECU **105a** is no longer operational. In the first alternative, steps **306-308** are performed. In the second alternative, steps **310-312** are performed.

(53) In the step **306**, the service center **130** requests the third data block **200** from the original first ECU **105a**. The original first ECU **105a** may still be on board the vehicle **100** or may be removed from the vehicle **100** and attached to equipment of the service center **130**. Next, in the step **308**, the original first ECU **105a** encrypts and transmits the third data block **200** to the service center **130**.

(54) In the step **310**, the service center **130** requests the second data block **225** from the second ECU **110**. Next, in the step **312**, the second ECU **110** transmits the second data block **225** to the service center **130**.

(55) After the alternative box **304**, in a step **314**, the service center **130** transmits either the second data block **225** or the third data block **200** to the remote server **125**. The service center **130** may include the VIN of the vehicle **100** in the transmission.

(56) Next, in a step **316**, the remote server **125** decrypts the third data block **200** or second data block **225**, as described above.

(57) Next, an alternative box **318** include a first alternative, that the decryption failed, and a second alternative, that the decryption succeeded. In the first alternative, a step **320** is performed.

(58) In the second alternative, steps **322-364** are performed, i.e., the rest of the sequence **300**.

(59) In the step **320**, the remote server **125** transmits a message to the service center **130** indicating that the decryption failed. The sequence **300** ends.

(60) In the step **322**, the remote server **125** generates and encrypts the first data block **250**, as described above. Next, in a step **324**, the remote server **125** transmits the first data block **250** to the service center **130**.

(61) Next, in a step **326**, the service center **130** transmits the first data block **250** to the replacement first ECU **105b**. The first data block **250** may remain encrypted. The replacement first ECU **105b** may or may not yet be installed into the vehicle **100**.

(62) Next, in a step **328**, the replacement first ECU **105b** decrypts the first data block **250**.

(63) Next, in a step **330**, the replacement first ECU **105b** transmits an acknowledgement indicating that the decryption was successful or a negative acknowledgement indicating that the decryption was unsuccessful.

(64) Next, in a step **332**, the replacement first ECU **105b** determines the VIN from the vehicle network **115**. Transmissions over the vehicle network **115** from other ECUs may include the VIN. From this point through the rest of the sequence **300**, the replacement first ECU **105b** is installed on board the vehicle **100**.

(65) Next, in a step **334**, the replacement first ECU **105b** requests the second data block **225** from the second ECU **110** over the vehicle network **115**. Next, in a step **336**, the second ECU **110** transmits the second data block **225** to the replacement first ECU **105b** over the vehicle network **115**, and the replacement first ECU **105b** receives the second data block **225**.

(66) Next, in a step **338**, the replacement first ECU **105b** decrypts the second data block **225**.

(67) Next, in a step **340**, the replacement first ECU **105b** compares the first data block **250** and the second data block **225**, as described above.

(68) Next, an alternative box **342** includes a first alternative and a second alternative. The first

alternative occurs upon the decryption failing in the step **338** or upon the replacement first ECU **105b** identifying a mismatch between the first data block **250** and the second data block **225** in the step **340**, as described above. The first alternative may also occur upon the first VIN **260** or the second VIN **235** not matching the VIN received in the step **332** above. The second alternative occurs upon the decryption being successful and the replacement first ECU **105b** identifying a match between the first data block **250** and the second data block **225**. In the first alternative, steps **344-362** are performed. In the second alternative, a step **364** is performed.

(69) In the step **344**, the replacement first ECU **105b** outputs a request for a fourth data block to the service center **130**, as described above. The fourth data block was previously downloaded from the vehicle **100** in the step **308** or the step **312** above.

(70) Next, an alternative box **346** includes a first alternative, that the replacement first ECU **105b** received the fourth data block from the service center **130**, and a second alternative, that the replacement first ECU **105b** failed to receive the fourth data block from the service center **130**. In the first alternative, steps **348-360** are performed. In the second alternative, a step **362** is performed.

(71) In the step **348**, the replacement first ECU **105b** decrypts the fourth data block, as described above. Next, in a step **350**, the replacement first ECU **105b** compares the first data block **250** and the fourth data block, e.g., the first security feature **270** and the fourth security feature, as described above.

(72) Next, an alternative box **352** includes a first alternative, that the replacement first ECU **105b** identified a second mismatch between the first data block **250** and the fourth data block, and a second alternative, that the replacement first ECU **105b** identified a second match between the first data block **250** and the fourth data block. In the first alternative, a step **354** is performed. In the second alternative, steps **356-360** are performed.

(73) In the step **354**, the replacement first ECU **105b** maintains the stored odometer value **205** on the replacement first ECU **105b** and instructs a component of the vehicle **100** to disable an operation of the component, as described above. The replacement first ECU **105b** may also log the third data block **200** with the unchanged stored odometer value **205** with the remote server **125** by encrypting and transmitting the third data block **200** to the remote server **125**. After the step **354**, the sequence **300** ends.

(74) In the step **356**, the replacement first ECU **105b** replaces the stored odometer value **205** on the replacement first ECU **105b** with the first odometer value **255**, as described above. Next, in a step **358**, the replacement first ECU **105b** logs the first data block **250** with the remote server **125** by encrypting and transmitting the third data block **200**, generated after the first odometer value **255** replaced the stored odometer value **205**, to the remote server **125**. Next, in a step **360**, the replacement first ECU **105b** encrypts and transmits the third data block **200** to the second ECU **110**, as described above. This is the first transmission as part of the periodic transmission of the third data block **200** to the second ECU **110**. After the step **360**, the sequence **300** ends.

(75) In the step **362**, the replacement first ECU **105b** maintains the stored odometer value **205** on the replacement first ECU **105b** and instructs a component of the vehicle **100** to disable an operation of the component, as described above. The replacement first ECU **105b** may also log the third data block **200** with the unchanged stored odometer value **205** with the remote server **125** by encrypting and transmitting the third data block **200** to the remote server **125**. After the step **362**, the sequence **300** ends.

(76) In the step **364**, the replacement first ECU **105b** encrypts and transmits the third data block **200** to the second ECU **110**, as described above. This is the first transmission as part of the periodic transmission of the third data block **200** to the second ECU **110**. After the step **364**, the sequence **300** ends.

(77) FIG. **4** is a process flow diagram illustrating an example process **400** for the first ECU **105** to facilitate replacement of the first ECU **105**. The memory of the first ECU **105** stores executable

instructions for performing the steps of the process **400** and/or programming can be implemented in structures such as mentioned above. As a general overview of the process **400**, the first ECU **105** increments the stored odometer value **205** and periodically transmits the third data block **200** to the second ECU **110**. Upon receiving a request for the third data block **200** from the service center **130**, the first ECU **105** transmits the third data block **200** to the service center **130**.

(78) Next, upon receiving the first data block **250** from the remote server **125**, the first ECU **105** determines the VIN, requests and receives the second data block **225** from the second ECU **110**, requests the fourth data block from the service center **130** upon identifying a mismatch between the first data block **250** and the second data block **225**, and replaces the stored odometer value **205** with the first odometer value **255** upon identifying a match between the first data block **250** and the second data block **225** or upon identifying a second match between the first data block **250** and the fourth data block. Upon identifying a second mismatch between the first data block **250** and the fourth data block, the first ECU **105** maintains the stored odometer value **205** and instructs a component to disable an operation of the component. Finally, the first ECU **105** transmits an update indicating whether the first odometer value **255** replaced the stored odometer value **205**.

(79) The process **400** begins in a block **405**, in which the first ECU **105** increments the stored odometer value **205** with distance traveled by the vehicle **100**, as described above.

(80) Next, in a decision block **410**, the first ECU **105** determines whether the period has elapsed since a previous transmission of the third data block **200** to the second ECU **110**, e.g., whether the vehicle **100** has traveled the preset distance since the previous transmission. Upon the period elapsing, the process **400** proceeds to a block **415**. Otherwise, the process **400** proceeds to a decision block **420**.

(81) In the block **415**, the first ECU **105** encrypts and transmits the third data block **200** to the second ECU **110**, as described above. After the block **415**, the process **400** proceeds to the decision block **420**.

(82) In the decision block **420**, the first ECU **105** determines whether the service center **130** requested the third data block **200**. If so, the process **400** proceeds to a block **425**. If not, the process **400** returns to the block **405** to continue incrementing the stored odometer value **205**.

(83) In the block **425**, the first ECU **105** encrypts and transmits the third data block **200** to the service center **130**, as described above.

(84) Next, in a decision block **430**, the first ECU **105** determines whether the first ECU **105** received the first data block **250** from the remote server **125**. Upon receiving the first data block **250** from the remote server **125**, the process **400** proceeds to a block **435**. Otherwise, the process **400** returns to the block **405** to continue incrementing the stored odometer value **205**.

(85) In the block **435**, the first ECU **105** determines the VIN from the vehicle network **115**, as described above.

(86) Next, in a block **440**, the first ECU **105** requests and receives the second data block **225** from the second ECU **110** while the first ECU **105** is installed on board the vehicle **100**, e.g., over the vehicle network **115**, as described above.

(87) Next, in a decision block **445**, the first ECU **105** compares the first data block **250** and the second data block **225**, as described above. Upon identifying a match between the first data block **250** and the second data block **225**, the process **400** proceeds to a block **450**. Upon identifying a mismatch between the first data block **250** and the second data block **225** or upon the first VIN **260** or the second VIN **235** not matching the VIN determined in the block **435**, the process **400** proceeds to a block **455**.

(88) In the block **450**, the first ECU **105** replaces the stored odometer value **205** on the first ECU **105** with the first odometer value **255**, as described above. After the block **450**, the process **400** proceeds to a block **475**.

(89) In the block **455**, the first ECU **105** outputs a request for the fourth data block to the service center **130**, as described above. The fourth data block was previously downloaded from the vehicle

100, either as the third data block **200** in the block **425** or as the second data block **225** from the second ECU **110**.

(90) Next, in a decision block **460**, the first ECU **105** compares the first data block **250** and the fourth data block, as described above. Upon identifying a second match between the first data block **250** and the fourth data block, the process **400** proceeds to the block **450**. Upon identifying a second mismatch between the first data block **250** and the fourth data block or upon failing to receive the fourth data block, the process **400** proceeds to a block **465**.

(91) In the block **465**, the first ECU **105** maintains the stored odometer value **205** on the first ECU **105**, as described above.

(92) Next, in a block **470**, the first ECU **105** instructs a component of the vehicle **100** to disable an operation of the component, as described above. After the block **470**, the process **400** proceeds to the block **475**.

(93) In the block **475**, the first ECU **105** logs the third data block **200** with the remote server **125**, either with the stored odometer value **205** replaced by the first odometer or not, as described above. After the block **475**, the process **400** ends.

(94) FIG. 5 is a process flow diagram illustrating an example process **500** for the second ECU **110** to replace the first ECU **105**. The memory of the second ECU **110** stores executable instructions for performing the steps of the process **500** and/or programming can be implemented in structures such as mentioned above. As a general overview of the process **500**, upon receiving the third data block **200**, the second ECU **110** stores the third data block **200**, which is thereby referred to as the second data block **225**. Upon receiving a request for the second data block **225**, the second ECU **110** transmits the second data block **225** to the requester.

(95) The process **500** begins in a decision block **505**, in which the second ECU **110** determines whether the second ECU **110** has received the third data block **200** from the first ECU **105**. Upon receiving the third data block **200**, the process **500** proceeds to a block **510**. If the second ECU **110** has not received a new third data block **200**, the process **500** proceeds to a decision block **515**.

(96) In the block **510**, the second ECU **110** stores the third data block **200**, which is now referred to as the second data block **225**. The second data block **225** is already encrypted. The second ECU **110** does not decrypt the second data block **225** and may not be capable of decrypting the second data block **225**, i.e., may lack the decryption key. After the block **510**, the process **500** proceeds to the decision block **515**.

(97) In the decision block **515**, the second ECU **110** determines whether a request has been received for the second data block **225**. The request may originate from the first ECU **105**, as described above with respect to the block **440** of the process **400** and the steps **334-336** of the sequence **300**. Alternatively, the request may originate from the service center **130**, as described above with respect to the steps **310-312** of the sequence **300**. Upon receiving the request for the second data block **225**, the process **500** proceeds to a block **520**. If no request has been received, the process **500** ends.

(98) In the block **520**, the second ECU **110** transmits the second data block **225** to the requester, either the first ECU **105** or the service center **130**. After the block **520**, the process **500** ends.

(99) FIG. 6 is a process flow diagram illustrating an example process **600** for the remote server **125** to replace the first ECU **105** in the vehicle **100**. The memory of the remote server **125** stores executable instructions for performing the steps of the process **600** and/or programming can be implemented in structures such as mentioned above. As a general overview of the process **600**, the remote server **125** receives the second data block **225** or the third data block **200** from the service center **130**. Upon successfully decrypting the third data block **200** or second data block **225**, the remote server **125** generates the first data block **250** and transmits the first data block **250** to the first ECU **105**. Upon unsuccessfully attempting to decrypt the third data block **200** or second data block **225**, the remote server **125** requests that the service center **130** retransmit the third data block **200** or second data block **225**.

(100) The process **600** begins in a block **605**, in which the remote server **125** receives the third data block **200** or second data block **225**, as described above. The service center **130** transmits the third data block **200** if the original first ECU **105a** is operational and transmits the second data block **225** if the original first ECU **105a** is not operational.

(101) Next, in a decision block **610**, the remote server **125** decrypts the third data block **200** or second data block **225**. Upon unsuccessfully attempting to decrypt the third data block **200** or second data block **225**, the process **600** proceeds to a block **615**. Upon successfully decrypting the third data block **200** or second data block **225**, the process **600** proceeds to a block **620**.

(102) In the block **615**, the remote server **125** transmits a request to the service center **130** to retransmit the third data block **200** or second data block **225**. After the block **615**, the process **600** ends.

(103) In the block **620**, the remote server **125** generates the first data block **250**, as described above.

(104) Next, in a block **625**, the remote server **125** transmits the first data block **250** to the first ECU **105**, as described above. After the block **625**, the process **600** ends.

(105) In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an on-board vehicle computer, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

(106) Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

(107) A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

(108) Databases, data repositories or other data stores described herein may include various kinds

of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a nonrelational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

(109) In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

(110) In the drawings, the same reference numbers indicate the same elements. Further, some or all of these elements could be changed. With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted.

(111) All terms used in the claims are intended to be given their plain and ordinary meanings as understood by those skilled in the art unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. The adjectives “first,” “second,” “third,” and “fourth” are used throughout this document as identifiers and are not intended to signify importance, order, or quantity. Use of “in response to,” “upon determining,” “upon receiving,” etc. indicates a causal relationship, not merely a temporal relationship.

(112) The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A computer comprising a processor and a memory, the memory storing instructions executable by the processor to: receive a first data block from a server remote from the computer and remote from a vehicle, the first data block including a first odometer value; once the computer is installed as a replacement on board the vehicle, receive a second data block from an electronic control unit (ECU) on the vehicle, the second data block including a second odometer value; compare the first data block and the second data block; upon identifying a match between the first data block and the second data block, replace a stored odometer value on the computer with the first odometer value; and upon identifying a mismatch between the first data block and the second data block, maintain the stored odometer value on the computer.
2. The computer of claim 1, wherein the instructions further include instructions to increment the stored odometer value with distance traveled by the vehicle.
3. The computer of claim 1, wherein the instructions further include instructions to transmit a third data block to the ECU, the third data block including the stored odometer value.
4. The computer of claim 3, wherein the instructions further include instructions to periodically

transmit the third data block to the ECU.

5. The computer of claim 3, wherein the instructions further include instructions to, upon the vehicle traveling a preset distance since a previous transmission of the third data block to the ECU, transmit the third data block to the ECU.

6. The computer of claim 3, wherein the instructions further include instructions to calculate a security feature and apply the security feature to the third data block before transmitting the third data block to the ECU.

7. The computer of claim 1, wherein the first odometer value is less than the stored odometer value.

8. The computer of claim 1, wherein the match is an equality between the first odometer value and the second odometer value, and the mismatch is an inequality between the first odometer value and the second odometer value.

9. The computer of claim 1, wherein the first data block includes a first vehicle identification number (VIN), the second data block includes a second VIN, the match is an equality between the first VIN and the second VIN, and the mismatch is an inequality between the first VIN and the second VIN.

10. The computer of claim 9, wherein the instructions further include instructions to, upon the first VIN not matching a third VIN of the vehicle, maintain the stored odometer value on the computer.

11. The computer of claim 10, wherein the instructions further include instructions to determine the third VIN from a vehicle network of the vehicle.

12. The computer of claim 9, wherein the instructions further include instructions to, upon the second VIN not matching a third VIN of the vehicle, maintain the stored odometer value on the computer.

13. The computer of claim 1, wherein the instructions further include instructions to, upon identifying the mismatch between the first data block and the second data block, instruct a component of the vehicle to disable an operation of the component.

14. The computer of claim 1, wherein the instructions further include instructions to, upon identifying the mismatch between the first data block and the second data block, output a request for a fourth data block, the fourth data block previously downloaded from the vehicle, the fourth data block including a fourth odometer value.

15. The computer of claim 14, wherein the instructions further include instructions to: compare the first data block and the fourth data block; upon identifying a second match between the first data block and the fourth data block, replace the stored odometer value on the computer with the first odometer value; and upon identifying a second mismatch between the first data block and the fourth data block, maintain the stored odometer value on the computer.

16. The computer of claim 15, wherein the first data block includes a first security feature, the fourth data block includes a fourth security feature, the second match is an equality between the first security feature and the fourth security feature, and the second mismatch is an inequality between the first security feature and the fourth security feature.

17. The computer of claim 16, wherein the first security feature is calculated from the first odometer value, and the fourth security feature is calculated from the fourth odometer value.

18. The computer of claim 14, wherein the instructions further include instructions to, upon failing to receive the fourth data block, maintain the stored odometer value on the computer.

19. The computer of claim 1, wherein the instructions further include instructions to decrypt the first data block, and decrypt the second data block.

20. A method comprising: receiving a first data block from a server remote from a computer and remote from a vehicle, the first data block including a first odometer value; receiving a second data block from an electronic control unit (ECU) on the vehicle by the computer, the second data block including a second odometer value, the computer receiving the second data block from the ECU once the computer is installed as a replacement on board the vehicle; comparing the first data block and the second data block; upon identifying a match between the first data block and the second

data block, replacing a stored odometer value on the computer with the first odometer value; and upon identifying a mismatch between the first data block and the second data block, maintaining the stored odometer value on the computer.
